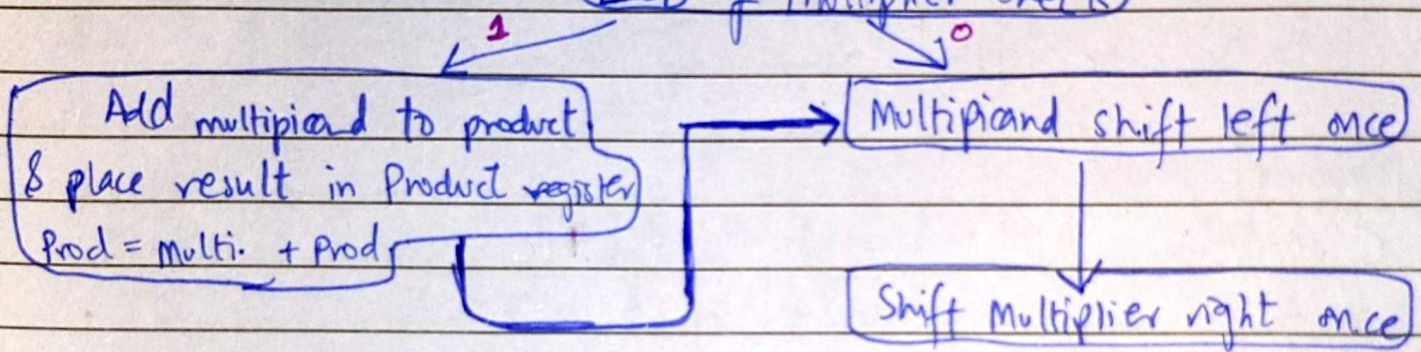


$\Rightarrow$ 3.3 & 3.4 from Book than Sides:

LSB of Multiplier Check



extra needed bcs we need to left shift

Example: $\begin{array}{r} 0000\ 0010 \rightarrow \text{multiplicand} \\ \times \quad \quad 0011 \rightarrow \text{multiplier} \end{array}$

Iter.	Steps	Multiplier	Multiplicand	Product
0	1a: Prod = Prod + multiplicand Initial vals.	0100	0000 1011	0000 0000
1	1a: Prod = prod + multiplicand 1b: shift multiplicand left once 1c: multiplier right-shift once	0010	0000 10110	0000 1011
2	2b: Shift multiplicand left once 2c: Multiplier right shift once	No step 'a' as zero 0000	0010 1100	0000 1011
3	3a: Prod = prod + multiplicand 3b: " 3c: "	0000	0101 1000	0011 0111
	since all zeros so we'll stop			0011 0111 answer.

$$\begin{matrix} ++ \\ -- \end{matrix} \} +ve \quad \begin{matrix} +- \\ -+ \end{matrix} \} -ve$$

Date _____

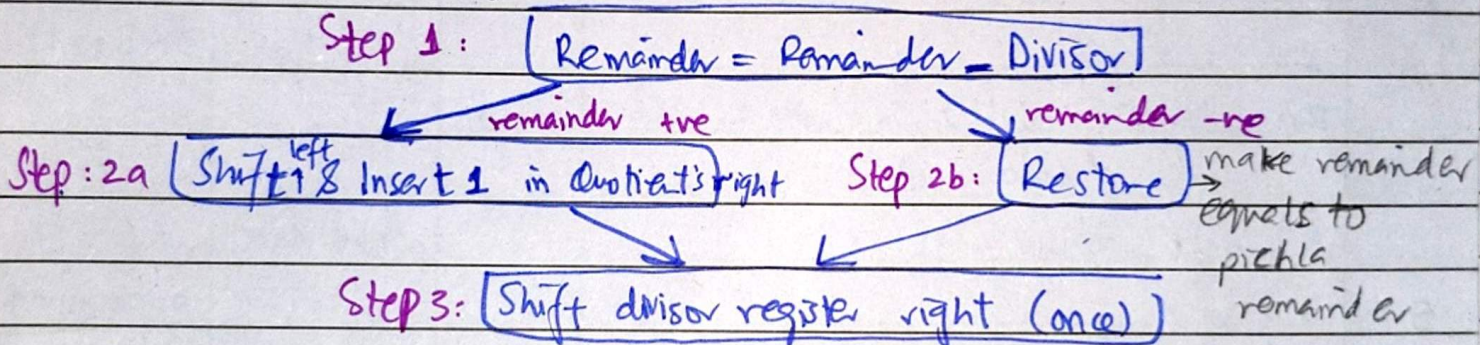
mul

⇒ **Signed Bit**: If multiplicand & multiplier have opposite signs; we'll preserve the sign bit in the and
Rest all the process of multiplying remains same for magnitude

□ **Division**:

$$\begin{array}{r} \text{divisor} \leftarrow 1000 \quad \begin{array}{c} \textcircled{1} \textcircled{4} \textcircled{5} \textcircled{6} \\ 1001 \end{array} \\ \hline 100 \ 1010 \rightarrow \text{dividend} \geq \text{divisor} \\ - 1000 \downarrow \downarrow \downarrow \\ \hline \textcircled{2} \textcircled{3} \textcircled{4} \textcircled{5} \\ 1010 \\ - \textcircled{6} 1000 \\ \hline \textcircled{7} 10 \rightarrow \text{remainder} \end{array}$$

⇒ **Flow Chart**:



next
page

Q: $(7)_{10} \div (2)_{10} = (0111)_2 \div (0010)_2$

Divisor (2) should be in higher (left) bits as we need to keep shifting

Date

initialised with dividend (7)

Iter.	Step	Quotient	Divisor	Remainder
0	Initial values	0000	0010 0000	0000 0111
1	1: rem = rem - Divisor 2b: Restore (as remainder is -ve) so bring old remainder (0000 0111) back 3: Shift divisor right (once)			1110 0111 0000 0111
2	1: rem = rem - Divisor 2b: restore as (1110 0111) is -ve 3: Shift divisor right		0001 0000	1111 0111 0000 0111
3	1: " 2b: " 3: "		0000 1000	1111 1111 0000 0111
4	1: " 2a: Shift ^{left} 18 insert 1 in Quotient's right since (0000 0111) is +ve 3: Shift divisor right	0001	0000 0100	0000 0011
5	1: " 2a: Shift ^{left} 8 insert 1 in Quotient's right 3: Shift divisor right	0011	0000 0010	0000 0001

* we'll stop on $n+1$ iteration where n is original no. of Bits of divisor.

Dividend = (Quotient $\times$ divisor) + remainder.